Example 5.2 (Illustrating SSS and SAS Congruence)

If $\angle E = \angle S$ and G is the midpoint of ES,

prove that $\Delta GET \equiv \Delta GST$.

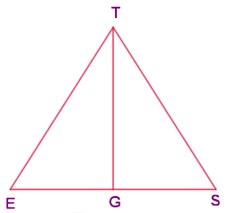
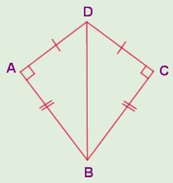


Fig. 5.8

Proof:

	Statements	Reasons
1	$\angle E \equiv \angle S$	given
2	$ET \equiv ST$	if angles, then sides
3	G is the midpoint of ES	given
4	$EG \equiv SG$	follows from 3
5	$TG \equiv TG$	common side (reflexive property)
6	$\Delta GET \equiv \Delta GST$	by SSS (2,4,5) & also by SAS (2,1,4)

Think



In the figure, $DA = DC$ and $BA = BC$.
Are the triangles DBA and DBC congruent?
Why?

Example 5.3 (Illustrating ASA Congruence)

If $\angle YTB \equiv \angle YBT$ and $\angle BOY \equiv \angle TRY$,

prove that $\Delta BOY \equiv \Delta TRY$

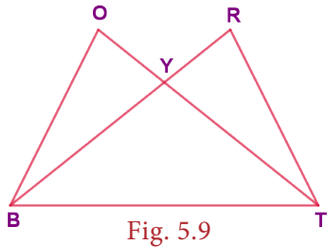


Fig. 5.9

Proof:

	Statements	Reasons
1	$\angle BYO \equiv \angle TYR$	vertical angles are congruent
2	$\angle YTB \equiv \angle YBT$	given
3	$BY \equiv TY$	if angles, then sides
4	$\angle BOY \equiv \angle TRY$	given
5	$\angle OBY \equiv \angle RTY$	follows from 1 and 4
6	$\Delta BOY \equiv \Delta$TRY	by ASA (1,3,5)

Example 5.4 (Illustrating RHS Congruence)

If TAP is an isosceles triangle with $TA = TP$ and $\angle TSA = 90°$.

(i) Is $\triangle TAS \equiv \triangle TPS$? Why?

(ii) Is $\angle P = \angle A$? Why?

(iii) Is $AS = PS$? Why?

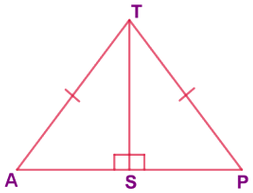


Fig. 5.10

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